



CS35 I: HUMAN COMPUTER INTERACTION

Group number: 2





Topic: GYM Management Application

Project Name: FitRout

Plan Your Route. Reach Your Goal.

All phases

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Project proposal

Project Problem or idea:

Many gyms still manage memberships, attendance, and workout schedules using fragmented tools such as spreadsheets, paper records, or messaging apps. This often causes membership status confusion, slow check-ins during peak hours, and lost or inconsistent workout plans shared through informal channels. This project proposes FitRoute, a gym management application that integrates membership management, workout scheduling, and attendance tracking into one unified system with role-based interfaces for admins, receptionists, trainers, and members.

Project Purpose:

The primary purpose of this project is to develop a Gym Management Application that digitizes the fitness experience. The system is designed to streamline administrative tasks and provide members with a modern, user-friendly interface to manage their healthy journey.

General expectations of the system:

Centralized Management: To store and manage member profiles, subscriptions, and payment records in one secure place.

Enhanced Accessibility: To allow members to book classes, reserve trainers, and track their schedules anytime via their smartphones

Operational Efficiency: To reduce manual paperwork for gym staff and provide automated reminders for membership renewals



Analysis:

Many gyms still manage daily work using paper notes or Excel sheets because it feels fast and familiar, especially during busy hours. The result is scattered information, missed expiry dates, slower check-ins, and outdated workout plans.

FitRoute brings membership, attendance, and workout schedules into one system, so everyone sees the same updated data.

Our system should also keep what already works in existing systems like, quick front-desk service, simple steps, and clear “yes/no” decisions that reduce confusion. It should support QR/barcode check-in to reduce typing and errors and provide workout templates (Push/Pull/Legs, beginner, strength) with easy editing instead of creating plans from zero.

Existing systems like Mindbody, Zen Planner, and Glofox show that these ideas work. FitRoute will apply them with a lighter, more direct interface designed for daily gym use.

The Users:

Primary and secondary users

Primary users

1- Receptionist:

frequent daily use and work under time pressure. Tasks include searching for members, check-in, confirming membership status (active/expired/frozen), and renewing or selling memberships.

2- Gym Member (Customer):

uses the system mainly on mobile to view membership details (plan and expiry date), attendance history, and workout schedules.

Secondary users

1- Coach:

creates, edits, and assigns workout schedules, and updates plans based on progress.

2- Owner:

uses the system less frequently to monitor performance through summaries such as active members, expiring memberships, attendance trends, and peak hours.

Persona:



Name: Sara
Age: 25
Role: Receptionist at FitRoute Gym
Tech Level: Intermediate
Device Used: Desktop at front desk

Bio:
Sara manages member check-ins, membership renewals, and customer inquiries during busy hours

Goals:

- Complete check-ins quickly
- Confirm membership status instantly
- Reduce waiting time at the desk

Usage Frequency

- **Daily**
- Weekly
- Occasionally

Pine point

- Slow systems during peak hours
- Manual data entry errors

Motivations:

- Provide fast and professional service
- Avoid mistakes in member records



Name: Ahmad
Age: 22
Role: University Student / Gym Member
Tech Level: Comfortable with mobile apps
Device Used: Mobile phone

Bio:
Ahmad goes to the gym regularly and wants an easy way to track his workouts and membership details.

Goals:

- Check membership expiry date
- View workout schedule
- Track attendance history

Usage Frequency

- Daily
- **Weekly**
- Occasionally

Pine point

- Forgetting workout plans
- Not knowing membership status clearly

Motivations:

- Stay consistent with workouts
- Improve fitness performance



Name: Khalid
Age: 38
Role: Gym Owner
Tech Level: Basic to Intermediate
Device Used: Desktop / Mobile

Bio:
Khalid oversees gym operations and monitors overall performance.

Goals:

- Monitor active memberships
- View attendance trends
- Track peak hours

Usage Frequency

- Daily
- **Weekly**
- Occasionally

Pine point

- Lack of clear summaries
- Difficulty tracking overall performance

Motivations:

- Increase gym efficiency
- Improve business performance



The Tasks:

This analysis describes what users currently do and what the new system must support.

Current Situation

- Receptionists manually verify membership status.
- Members ask staff about schedules and expiry dates.
- Payments may require visiting the branch or using external websites.
- Coaches manage workout plans manually or through basic tools.
- Owners rely on manual summaries or fragmented reports.

System-Supported Tasks

Receptionist

- Search for members
- Check members in
- Verify membership status
- Renew membership
- Registering new members

Gym Member

- View membership details
- Check expiry date
- View attendance history
- Book and cancel classes
- Track workout progress

Coach

- Create workout plans
- Edit training programs
- Assign plans to members
- Update programs based on progress

Owner

- View active member summaries
- Monitor expiring memberships
- Review attendance statistics
- Analyze peak hours



The Questionnaire:

A total of 73 responses were collected.

1. Age

Most respondents are aged 18–22 (49.3%), followed by 23–30 (32.9%).

Analysis: The main target users are young adults, so the system should prioritize a modern and mobile-friendly design.

2. Gym Attendance

The majority attend the gym 2–3 times per week (34.2%).

Analysis: Since users visit regularly, the system must support quick and repeated interactions such as booking and check-in.

3. Gym App Usage

72.6% of respondents use a gym mobile application.

Analysis: There is strong demand for digital solutions, although some gyms still lack proper applications.

4. App Usage Frequency

Most users access the app several times per week (43.8%).

Analysis: Frequent use requires high usability and fast performance.

5. Desired Features

Top requested features include easy booking (52.1%), clear schedules (49.3%), workout tracking (47.9%), and a simple design (43.8%).

Analysis: Booking efficiency and schedule clarity are key priorities.

6. Reported Problems

Common issues include slow performance, complicated interfaces, too many buttons, payment redirection to websites, and unclear policies.

Analysis: Performance, simplicity, and integrated payment must be improved.

7. Importance Ratings

Over 50% rated ease of use, fast performance, minimal steps, and clear design as extremely important.

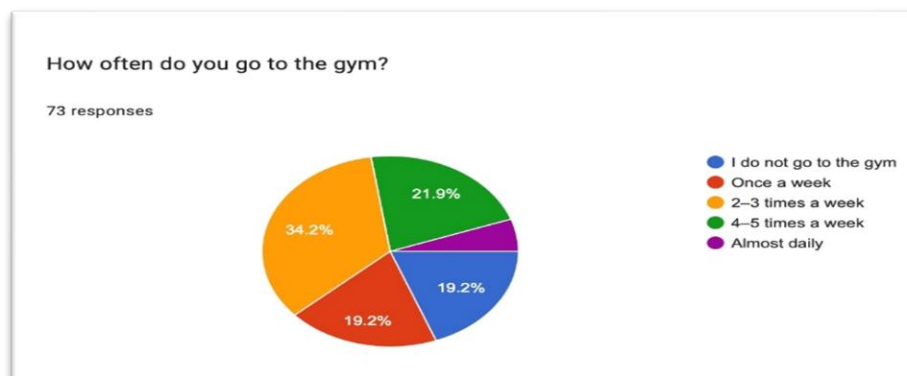
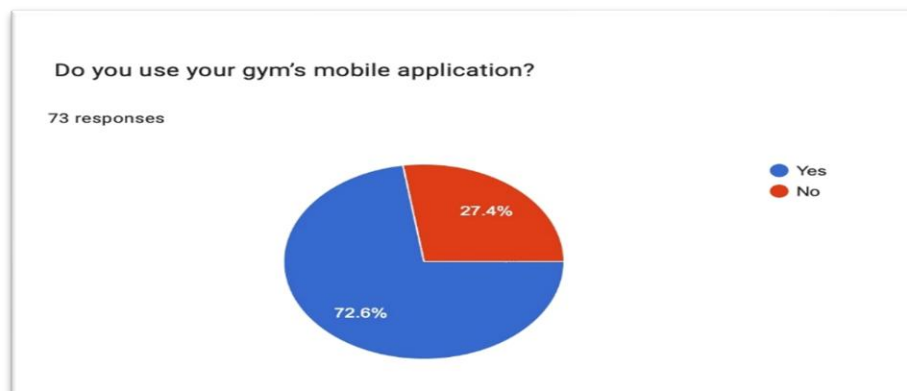
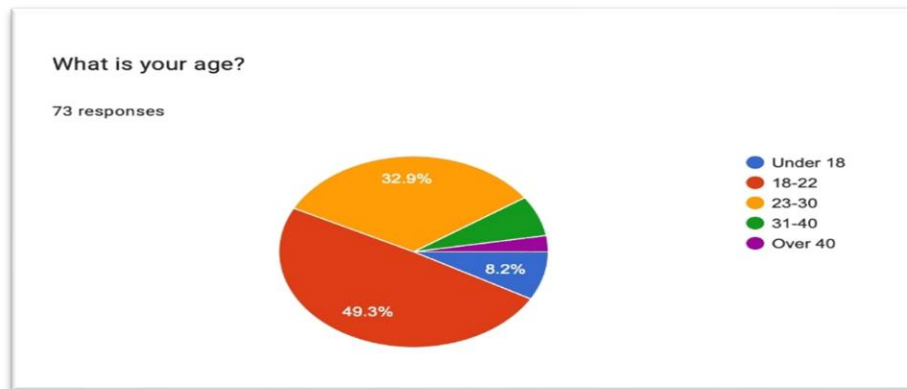
Analysis: Usability and speed are critical requirements for the new system.

8. Main Frustrations

Users reported booking difficulties, full classes, payment inconvenience, lack of notifications, slow applications, and unclear design.

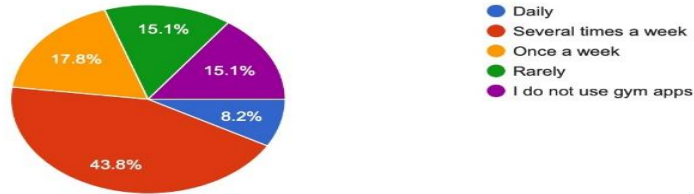
Analysis: The main issues relate to booking complexity, payment integration, performance problems, and interface clarity.

The Survey questions results



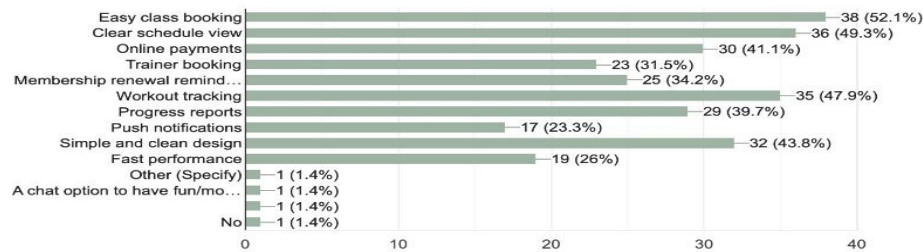
How often do you use the gym app?

73 responses



What features would you like a gym management app to include? (Select all that apply)

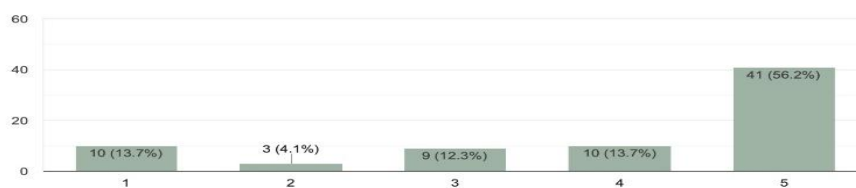
73 responses

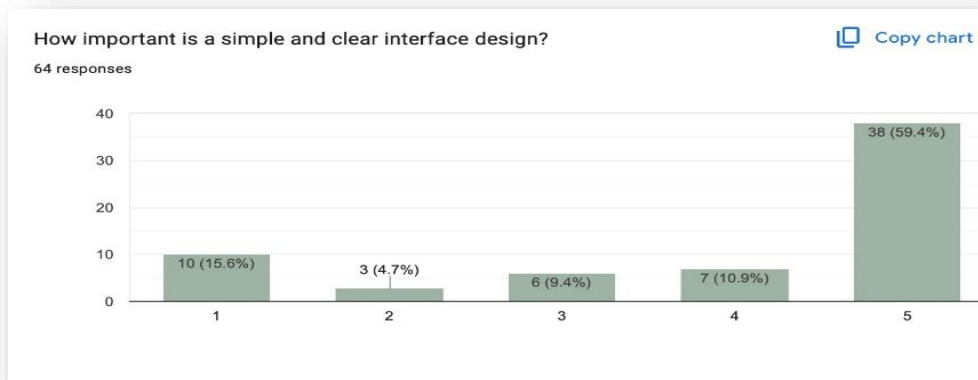
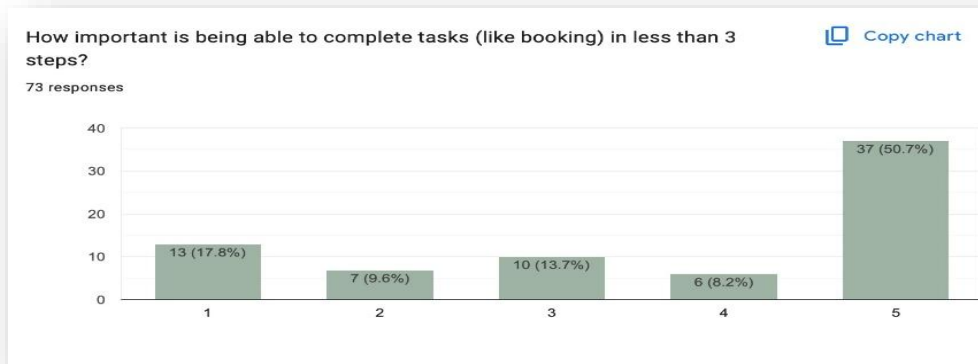
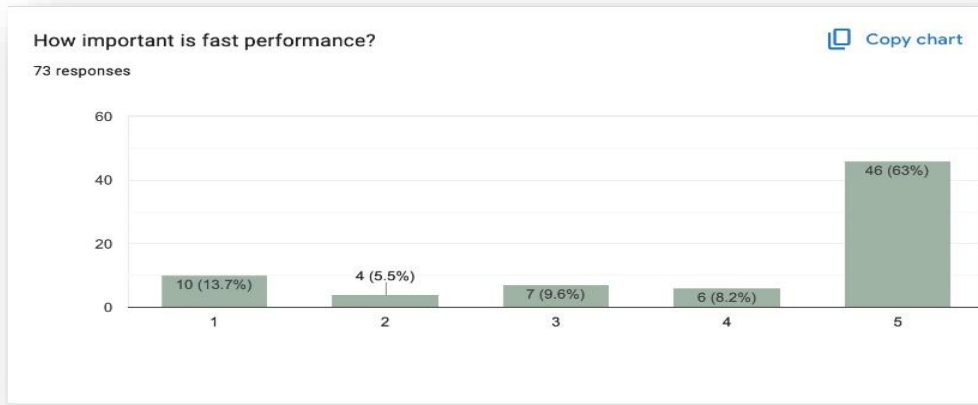


Linear Scale (1-5)

How important is ease of use in a gym app?

73 responses





What frustrates you most when dealing with gym management processes (booking, payments, schedules)?

73 responses

The Observation:

Based on reviewing existing gym applications and analyzing user feedback, it was observed that most users primarily utilize gym apps for booking classes, checking schedules, and managing their memberships. Ease of use and application performance are considered key factors influencing user satisfaction. Common issues reported by users include slow performance, complicated interfaces, and booking system errors. Additionally, users tend to discontinue using the application if it does not provide clear value or a smooth user experience from the beginning.

Requirements

The UI Functional Requirements

1. Users will be able to register for a new account and securely log in to the application.
2. Users will be able to access their membership details and complete membership renewals directly through the app.
3. Users will be able to view workout schedules and filter them by date, trainer, or class type.
4. Users will be able to monitor their attendance history within the application.
5. Users will receive timely notifications regarding membership expiration and upcoming class reminders.
6. Users will be able to reserve workout classes and check the number of available spots.
7. The system will enable trainers to modify and update class schedules as needed.
8. The system will provide users with a communication channel to reach trainers or customer support.

The Non-functional Requirements

1. The interface will be fully optimized to function smoothly on both mobile and tablet devices.
2. The system will ensure that all primary page's load within a maximum of three seconds to maintain efficient performance.
3. The application will maintain consistent visual design elements, including colors, icons, and typography across all screens.
4. The system will protect user data by applying secure privacy measures and safeguarding all personal information.
5. The interface will provide clear, informative error messages and confirmation prompts to guide users effectively.
6. The application will support bilingual usage, offering both Arabic and English language options.
7. The system will maintain high reliability, ensuring an uptime of at least 99%.
8. The design will adhere to accessibility standards, including readable font sizes, proper color contrast, and inclusive interface elements.

Logo Design:

The FitRoute logo represents the idea of guiding users on their fitness journey in a clear and simple way.

The most important element in the logo is the letter “R”, which is designed as a road, this road symbolizes the path that users follow to reach their fitness goals, such as improving health, tracking attendance, and following workout schedules. It reflects progress, direction, and consistency, which are key ideas in any fitness system.

The small dumbbell icon above the letter “i” represents the gym environment and physical training, it quickly shows that the application is related to fitness and exercise.

The green background color reflects health, growth, and energy, while the white text keeps the design clean and easy to read.

Overall, the logo presents FitRoute as a simple, modern, and user-friendly system that helps users stay on track with their fitness journey by combining memberships, attendance, and workout plans in one place.

FitRoute:



Colors:

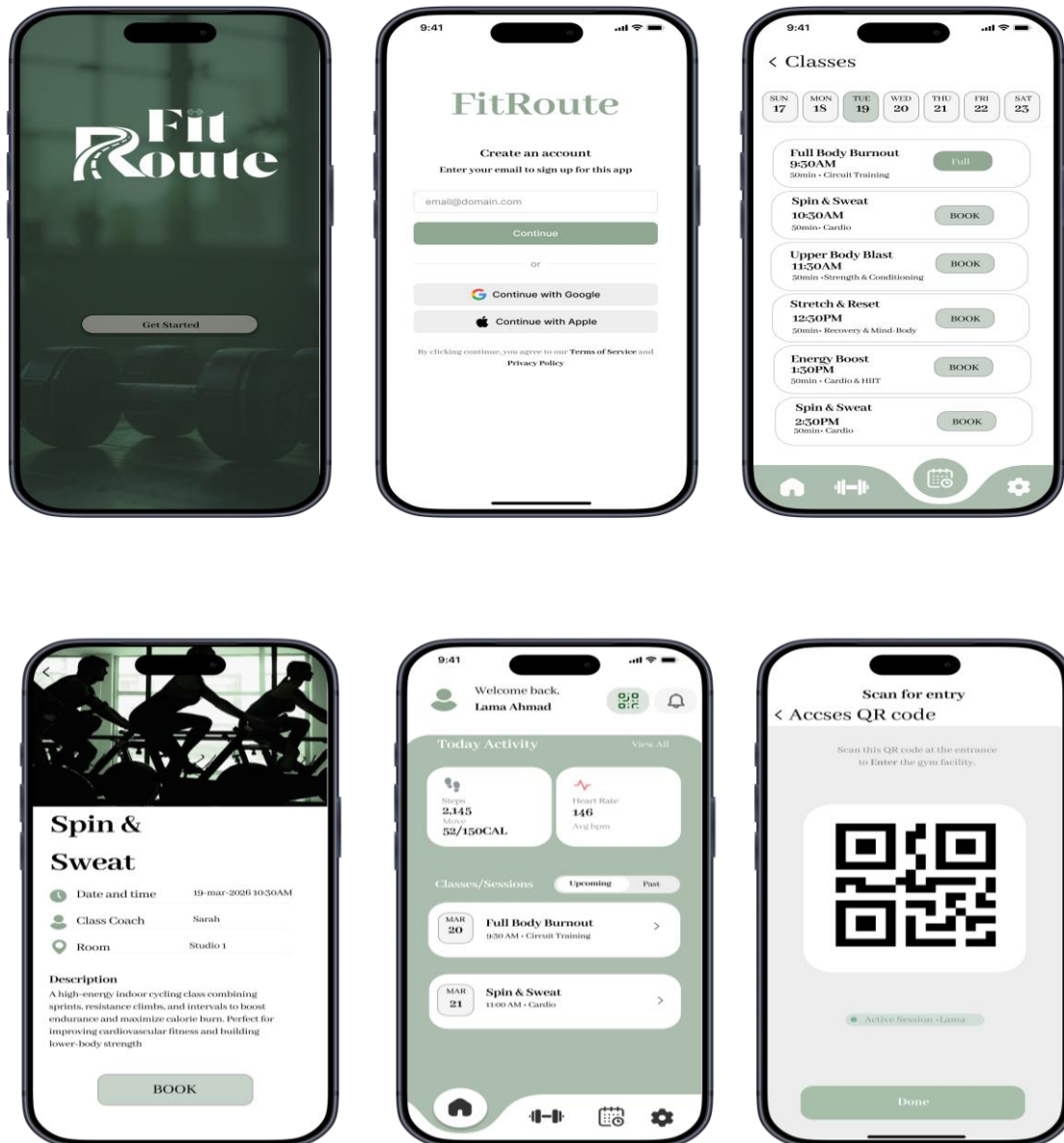


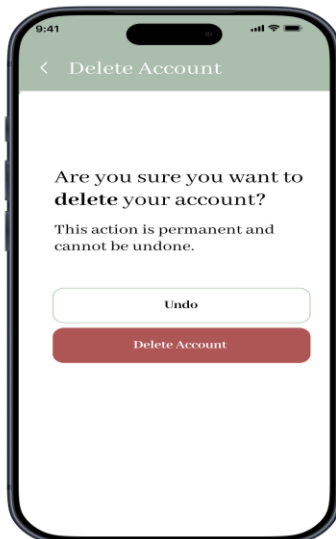
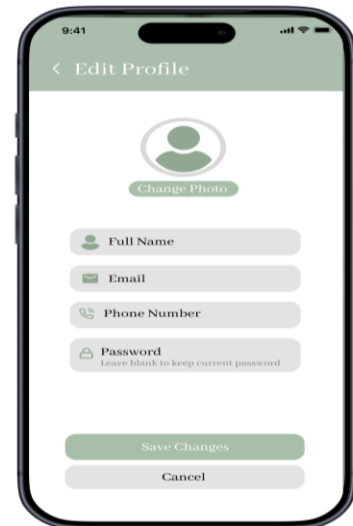
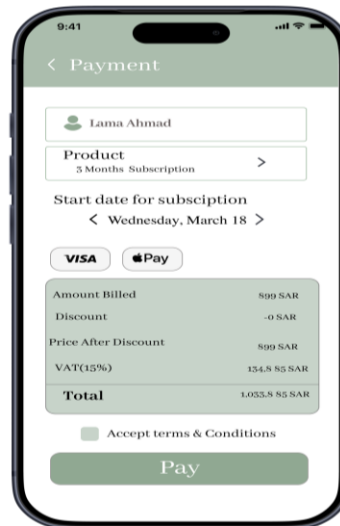
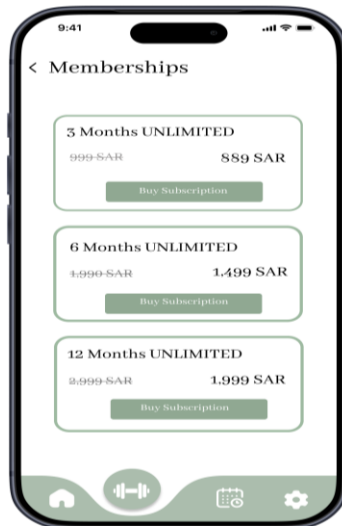
#335340

White

Prototype Design:

The FitRoute prototype was designed using Figma as a high-fidelity, clickable interface. It includes key screens such as login, dashboard, membership details, and class booking. The design focuses on simplicity, clear navigation, and fast user interaction to help users complete their tasks efficiently. [click here](#) to view it.







Key Tasks and Design:

Key Tasks:

1. Sign up / Log in
Create an account or sign in using email or social login.
2. View dashboard
Check steps, calories, heart rate, and upcoming sessions.
3. Browse class schedule
View classes by date, time, and availability.
4. Book a class
Select a session, view details, and confirm booking.
5. Access gym via QR code
Scan QR code for entry and confirm active session.
6. Manage membership and payments
View plans, subscription, and complete payments.
7. Manage profile and settings
Edit personal details, change language, or delete account.

Design Narrative:

The design was developed using a user-centered approach focusing on simplicity and efficiency. First, user needs were identified: quick booking, easy navigation, and clear information.

To address this the system follows a simple and intuitive user flow that minimizes steps and reduces user effort:

login → home dashboard → navigation to key features such as classes, membership, or settings → action (booking or payment).

The interface uses a clean layout with card-based components to visually separate content and improve information hierarchy such as stats, classes, and memberships. This improves readability and allows users to quickly scan content. Navigation is consistent through a bottom navigation bar, making it easy to switch between main sections without confusion. Key features such as QR code entry were added to reduce waiting time and improve user efficiency and a structured booking system were added to reduce time and effort for users.

The Interaction Style:

The design combines both menu-based and direct manipulation interaction styles. The bottom navigation bar allows users to quickly access the main sections of the application, such as the dashboard, classes, and profile.

At the same time, interactive elements such as buttons and cards enable users to perform actions directly, such as booking a class or viewing workout details. This reduces the number of steps required to complete tasks and improves overall usability.

This approach enhances the user experience by making the system intuitive, efficient, and easy to use, especially for frequent users who need quick and repeated access to core features.

Design Improvements:

- Faster task completion through simplified navigation and reduced steps.
- Reduced cognitive load through a clean, card-based layout and clear visual hierarchy.
- Improved usability with consistent navigation and readable design.
- Better user control through easy access to settings and profile management.
- Increased efficiency through QR code entry and quick access to key features from the dashboard.

Usability Evaluation:

The design has clear navigation, with a simple layout and easily recognizable icons; The use of color, font, and button style is consistent throughout every screen within the system making it easy for the user to find their way around the system; An easy and therefore intuitive layout makes it simple for new users to learn how to use the system; The number of steps needed to complete tasks within the system is minimal, making the total experience quick and user friendly.

Options that are clear and an organized layout of options reduces the possibility of error by the user. The modern clean look of the design overall provides the user with a positive user experience. Areas where improvements could be made are in the use of better spacing and smaller icon hints, as this will create a better overall usability.

Ai tool UX pilot

Strengths

- The overall interface is clean, modern, and visually consistent across all screens.
- Clear hierarchy makes it easy for users to understand actions (Book, Pay, Confirm, Save).
- Navigation icons are intuitive and support smooth movement between sections.
- Membership and payment screens are well-organized and easy to follow.
- Class details and activity tracking are presented in a structured, readable way.

Weaknesses

- Some screens contain dense information (especially the classes list), which may overwhelm new users.
- Slight variations in text size reduce visual consistency.
- The QR code screen has unused empty space that could be optimized.
- The language selection screen could benefit from more spacing between options.
- The delete account screen lacks strong visual warning cues.

Suggestions

- Add grouping or dividers to long lists to improve readability.
- Standardize font sizes across all screens.
- Improve spacing in the language screen for better clarity.
- Add warning colors or icons to the delete account screen.
- Use the empty space in the QR screen for brief instructions or a small visual element.



Category	Our Evaluation	AI Tool (UX Pilot)
Navigation & Layout	Clear navigation with simple layout and easy icons	Agrees – clear structure and intuitive navigation
Design Consistency	Consistent colors, fonts, and buttons	Agrees – visually consistent across screens
User Experience	Smooth, fast, and user-friendly experience	Agrees – clean, modern, and positive experience
Task Efficiency	Few steps to complete tasks	Agrees – quick and efficient process
Learning Ease	Easy to understand overall	Strong focus on being intuitive for new users
Content Density	Not strongly highlighted	Noticed some screens are too dense (e.g., classes list)
Font Consistency	Not mentioned	Noticed slight differences in font sizes
Spacing Issues	General improvement suggested	Highlighted specific spacing issues (QR & language screen)
Safety & Warnings	Noticing weak warning in deleted account screen	Not clearly emphasized



Usability testing:

User 1:

The first user interacted with the system smoothly. Most tasks were completed quickly, with only minor difficulties

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1. Sign up	2 min	0	Easy process	None
2. Sign in	1 min	0	No issues	None
3. View membership details	30 sec	0	Clear	None
4. Book a class	1.5 min	1	Slight delay in choosing class	Improve filtering
5. Cancel booking	30 sec	0	Easy	None
6. View class schedule	40 sec	0	Clear	None
7. QR check-in	15 sec	0	Very fast	None
8. View attendance	35 sec	0	Easy to use	None
9. Update profile	1 min	0	Smooth	None
10. Change language	20 sec	0	Simple	None

User 2:

The second user completed most tasks successfully.

The system was easy to use, and the interface was clear. However, the user faced some difficulty in updating the profile and booking a class. Overall, the system is usable but needs small improvements in some tasks.

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1. Sign up	2.5 min	1	Slight confusion in entering details	Make the fields more clear
2. Sign in	1 min	0	Easy to complete	None
3. View membership details	40 sec	0	Clear	None
4. Book a class	2 min	1	Needed more time to choose a class	Improve class organization
5. Cancel booking	40 sec	0	Easy	None
6. View class schedule	45 sec	0	Clear	None
7. QR check-in	20 sec	0	Fast and smooth	None
8. View attendance	40 sec	0	Easy to use	None
9. Update profile	1.5 min	1	User was confused when editing information	Improve button clarity
10. Change language	25 sec	0	Simple	None

User 3:

The third user completed all tasks successfully and in a shorter time compared to other users. The system was very easy to use, and the interface was clear and intuitive. The user did not face any difficulties while performing the tasks and was able to navigate the system smoothly. Overall, the system is highly usable and provides an efficient and seamless user experience.

Task	Time	Errors	Comments	Suggestion
1. Sign up	1.5 min	0	Very quick and clear	None
2. Sign in	45 sec	0	Fast	None
3. View membership details	25 sec	0	Very clear	None
4. Book a class	1.2 min	0	Smooth selection	None
5. Cancel booking	25 sec	0	Easy	None
6. View schedule	35 sec	0	Well organized	None
7. QR check-in	12 sec	0	Extremely fast	None
8. View attendance	30 sec	0	Simple and clear	None
9. Update profile	50 sec	0	Easy to edit	None
10. Change language	15 sec	0	Very simple	None

User 4:

The fourth user had an overall smooth and positive experience while using the system. All tasks were completed successfully and without difficulty, except for a minor issue during the attendance viewing task, which resulted in one error. The user was slightly confused while accessing the attendance details, but managed to complete the task in the end. Overall, the interface was clear, easy to navigate, and efficient, with only a small improvement needed in the attendance feature.

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1. Sign up	3 min	0	Easy and clear	None
2. Sign in	2 min	0	smooth	None
3. View membership details	30 sec	0	Easy to understand	None
4. Book a class	1.5 min	0	Smooth process	None
5. Cancel booking	30 sec	0	Easy	None
6. View class schedule	40 sec	0	Well Organized	None
7. QR check-in	15 sec	0	Very fast	None
8. View attendance	45 sec	1	Slight Confusion in accessing details	Improve Visibility of attendance
9. Update profile	1.50 min	0	Smooth	None
10. Change language	20 sec	0	Simple	None



User 5:

The fifth user completed most tasks successfully with a generally smooth experience. The interface was clear and easy to use, but the user had small difficulties in booking a class and updating profile. Overall, the system is usable but needs slight improvements for better clarity.

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1. Sign up	2 min	0	Easy and clear	None
2. Sign in	1 min	0	smooth	None
3. View membership details	35 sec	0	Easy to understand	None
4. Book a class	2 min	1	Took time to choose class	Improve filtering options
5. Cancel booking	35 sec	0	Easy	None
6. View class schedule	45 sec	0	Well Organized	None
7. QR check-in	18 sec	0	fast	None
8. View attendance	40 sec	0	Easy to use	None
9. Update profile	1.5 min	1	Slight Confusion in editing info	Make buttons more clear
10. Change language	20 sec	0	Simple	None



Usable or not:

FitRoute is considered usable because most participants were able to complete more than 80% of the tasks successfully, which meets the project requirements. This indicates that the system is effective and supports users in achieving their goals without major difficulties. The interface is simple, clear, and easy to navigate, allowing users to understand and use the features with minimal effort. Most tasks were completed smoothly and within a reasonable time, reflecting good system performance and overall design quality. Overall, the results show that FitRoute provides positive and efficient user experience

Enhancement:

The system can be improved by enhancing the class booking feature through better organization and filtering, making profile editing buttons clearer and more noticeable, and improving the visibility and accessibility of the attendance section for easier use.

Summary:

This comparison shows that evaluating user experience requires attention to both overall design and small details. The AI tool highlighted areas that were not initially noticed, such as spacing and content organization, while also confirming key strengths like clear navigation and consistency. Based on this, several improvements were identified to enhance clarity, usability, and user safety.



STUDENT NAME	STUDENT WORK
SHUJON ALI ALSHEHRI	○ Report- Project idea ○ Write key tasks and interaction style ○ Design three interfaces ○ Analysis ○ Usability test user 1
Hala Turki Alshahrani	○ UI Requirements ○ Usability evaluation & AI tools ○ Design three interfaces ○ Usability test user 2
Lana Mohammed Alyahya	○ Questionnaire- Tasks ○ Design Narrative & improvements ○ Design three interfaces ○ Usability test user 3
Lujain Mualla Alanazi	○ Analysis-Users (persona) ○ Logo design and description ○ Usability test user 4
Meead Eid Alqahtani	○ Project purpose -Observation ○ Comparison (evaluation with Ai tool) & summary ○ Design three interfaces ○ Usability test user5